

0902.0431: residuals

equations measured 2026-09-05T10:59:49Z against the model of 2026-09-02

Corrected (10)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00425 (was)	70	0.082	$\begin{aligned} & A \, X \, A^* \times A \, Y \, A^* = A(X \times Y) A^* \\ & (PMA^*)^* (PNA^*) = AM^* P^* PNA = A(M^* N) A, \\ & (PMA^*) (AYA^*) = P(MY) A^*, \\ & \overline{PMA^* \times PNA^*} = \overline{\tilde{t} \tilde{P}(M \times N)^t \tilde{A}^*} = \overline{PM \times NA^*}, \text{ etc.} \end{aligned}$	$\begin{aligned} & AXA^* \times AYA^* = A(X \times Y)A^* \\ & (PMA^*)^* (PNA^*) = AM^* P^* PNA = A(M^* N) A, \\ & (PMA^*) (AYA^*) = P(MY)A^*, \\ & \overline{PMA^* \times PNA^*} = \overline{\tilde{t} \tilde{P}(M \times N)^t \tilde{A}^*} = \overline{PM \times NA^*}, \text{ etc.} \end{aligned}$	—
0902.0431_E00425 (now)	70	—	$\begin{aligned} & A \, X \, A^* \times A \, Y \, A^* = A(X \times Y) A^* \\ & (PMA^*)^* (PNA^*) = AM^* P^* PNA = A(M^* N) A, \\ & (PMA^*) (AYA^*) = P(MY) A^*, \\ & \overline{PMA^* \times PNA^*} = \overline{\tilde{t} \tilde{P}(M \times N)^t \tilde{A}^*} = \overline{PM \times NA^*}, \text{ etc.} \end{aligned}$	$\begin{aligned} & AXA^* \times AYA^* = A(X \times Y)A^* \\ & (PMA^*)^* (PNA^*) = AM^* P^* PNA = A(M^* N) A, \\ & (PMA^*) (AYA^*) = P(MY)A^*, \\ & \overline{PMA^* \times PNA^*} = \overline{\tilde{t} \tilde{P}(M \times N)^t \tilde{A}^*} = \overline{PM \times NA^*}, \text{ etc.} \end{aligned}$	—
basis: inferred / ink					
0902.0431_E00425 (was)	70	0.082	$\begin{aligned} & A \, X \, A^* \times A \, Y \, A^* = A(X \times Y) A^* \\ & (PMA^*)^* (PNA^*) = AM^* P^* PNA = A(M^* N) A, \\ & (PMA^*) (AYA^*) = P(MY) A^*, \\ & \overline{PMA^* \times PNA^*} = \overline{\tilde{t} \tilde{P}(M \times N)^t \tilde{A}^*} = \overline{PM \times NA^*}, \text{ etc.} \end{aligned}$	$\begin{aligned} & AXA^* \times AYA^* = A(X \times Y)A^* \\ & (PMA^*)^* (PNA^*) = AM^* P^* PNA = A(M^* N) A, \\ & (PMA^*) (AYA^*) = P(MY)A^*, \\ & \overline{PMA^* \times PNA^*} = \overline{\tilde{t} \tilde{P}(M \times N)^t \tilde{A}^*} = \overline{PM \times NA^*}, \text{ etc.} \end{aligned}$	—
0902.0431_E00425 (now)	70	—	$\begin{aligned} & \nu \Big(\sum_{k=0}^3 \lambda_k H_k \Big) \\ & = \frac{1}{2} \Big(\lambda_0 (-H_0 + H_1 - H_2 + H_3) + \lambda_1 (-H_0 + H_1 + H_2 - \\ & \quad + \lambda_2 (H_0 + H_1 + H_2 + H_3) + \lambda_3 (-H_0 - H_1 + H_2 + H_3) \end{aligned}$	$\begin{aligned} & \lambda_0 - \lambda_1 = 1 \quad 0 \quad 0 \quad 0 \quad 0 \\ & \lambda_0 - \lambda_2 = 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_1 = 1 \quad 2 \quad \frac{1}{1} = \frac{1}{\lambda_1 - \lambda_2} = 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_3 = 1 \quad 1 \quad 2 \\ & \lambda_0 - \lambda_3 = 1 \quad 1 \quad 1 \quad 0 \quad 2 \end{aligned}$	—
basis: inferred / ink					
0902.0431_E00515 (was)	85	0.002	$\begin{aligned} & \lambda_0 - \lambda_1 = 1 \quad 0 \quad 0 \quad 0 \quad 0 \\ & \lambda_0 - \lambda_2 = 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_1 = 1 \quad 2 \quad \frac{1}{1} = \frac{1}{\lambda_1 - \lambda_2} = 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_3 = 1 \quad 1 \quad 2 \\ & \lambda_0 - \lambda_3 = 1 \quad 1 \quad 1 \quad 0 \quad 2 \end{aligned}$	$\begin{aligned} & \lambda_0 - \lambda_1 = 1 \quad 0 \quad 0 \quad 0 \quad 0 \\ & \lambda_0 - \lambda_2 = 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_1 = 1 \quad 2 \quad \frac{1}{1} = \frac{1}{\lambda_1 - \lambda_2} = 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_3 = 1 \quad 1 \quad 2 \\ & \lambda_0 - \lambda_3 = 1 \quad 1 \quad 1 \quad 0 \quad 2 \end{aligned}$	—
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00515 (now)	85	—	$\begin{aligned} \lambda_0 - \lambda_1 &= 1 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \lambda_0 - \lambda_2 &= 1 \ 1 \ 0 \ 0 \ 0 \ 0 \\ \lambda_0 - \lambda_3 &= 1 \ 1 \ 1 \ 0 \ 0 \ 0 \\ \lambda_1 - \lambda_2 &= 0 \ 1 \ 0 \ 0 \ 0 \ 0 \\ \lambda_1 - \lambda_3 &= 0 \ 1 \ 1 \ 0 \ 0 \ 0 \\ \lambda_2 - \lambda_3 &= 0 \ 0 \ 1 \ 0 \ 0 \ 0 \end{aligned}$	$\begin{aligned} \lambda_0 + \lambda_1 &= 1 \ 2 \ 2 \ 1 \ 1 \ 1 \\ \lambda_0 + \lambda_2 &= 1 \ 1 \ 2 \ 1 \ 1 \ 1 \\ \lambda_0 + \lambda_3 &= 1 \ 1 \ 1 \ 1 \ 1 \ 1 \\ \lambda_1 + \lambda_2 &= 0 \ 1 \ 2 \ 1 \ 1 \ 1 \\ \lambda_1 + \lambda_3 &= 0 \ 1 \ 1 \ 1 \ 1 \ 1 \\ \lambda_2 + \lambda_3 &= 0 \ 0 \ 1 \ 1 \ 1 \ 1 \end{aligned}$	—
basis: inferred / ink					
0902.0431_E00515 (was)	85	■0.002	$\begin{aligned} &\lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ &\lambda_1 - \lambda_2 = 1 \quad \lambda_1 - \lambda_3 = 1 \quad \lambda_2 - \lambda_3 = 1 \\ &\lambda_0 + \lambda_1 = 1 \quad \lambda_0 + \lambda_2 = 1 \quad \lambda_0 + \lambda_3 = 1 \\ &\lambda_1 + \lambda_2 = 1 \quad \lambda_1 + \lambda_3 = 1 \quad \lambda_2 + \lambda_3 = 1 \end{aligned}$	$\begin{aligned} \lambda_0 - \lambda_1 &= 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\ \lambda_0 - \lambda_2 &= 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \\ \lambda_0 - \lambda_3 &= 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 2 \end{aligned} \quad \frac{1}{1} = \frac{1}{\lambda_1 - \lambda_2} = 0 \quad \frac{1}{1} = \frac{1}{\lambda_1 - \lambda_2} = 0 \quad \frac{1}{1} = \frac{1}{\lambda_1 - \lambda_2} = 0$	—
0902.0431_E00515 (now)	85	—	$\begin{aligned} &\lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ &\lambda_1 - \lambda_2 = 1 \quad \lambda_1 - \lambda_3 = 1 \quad \lambda_2 - \lambda_3 = 1 \\ &\lambda_0 + \lambda_1 = 1 \quad \lambda_0 + \lambda_2 = 1 \quad \lambda_0 + \lambda_3 = 1 \\ &\lambda_1 + \lambda_2 = 1 \quad \lambda_1 + \lambda_3 = 1 \quad \lambda_2 + \lambda_3 = 1 \end{aligned}$	$\begin{aligned} \nu H &= \nu \left(\sum_{k=0}^3 \lambda_k H_k \right) \\ &= \frac{1}{2} (\lambda_0 (-H_0 + H_1 - H_2 + H_3) + \lambda_1 (-H_0 + H_1 + H_2 - H_3) \\ &\quad + \lambda_2 (H_0 + H_1 + H_2 + H_3) + \lambda_3 (-H_0 - H_1 + H_2 + H_3)) \\ &= \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) H_0 + \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1 \end{aligned}$	—
basis: inferred / ink					
0902.0431_E00756 (was)	125	■0.119	$\begin{aligned} &\lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ &\lambda_1 - \lambda_2 = 1 \quad \lambda_1 - \lambda_3 = 1 \quad \lambda_2 - \lambda_3 = 1 \\ &\lambda_0 + \lambda_1 = 1 \quad \lambda_0 + \lambda_2 = 1 \quad \lambda_0 + \lambda_3 = 1 \\ &\lambda_1 + \lambda_2 = 1 \quad \lambda_1 + \lambda_3 = 1 \quad \lambda_2 + \lambda_3 = 1 \end{aligned}$	$\begin{aligned} \lambda_0 - \lambda_1 &= 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\ \lambda_0 - \lambda_2 &= 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \\ \lambda_0 - \lambda_3 &= 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \\ \lambda_1 - \lambda_2 &= 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \\ \lambda_1 - \lambda_3 &= 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \\ \lambda_2 - \lambda_3 &= 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \end{aligned} \quad \begin{aligned} \lambda_0 + \lambda_1 &= 1 \quad 2 \quad 2 \quad 2 \quad 2 \quad 1 \\ \lambda_0 + \lambda_2 &= 1 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \\ \lambda_0 + \lambda_3 &= 1 \quad 1 \quad 1 \quad 2 \quad 2 \quad 1 \\ \lambda_1 + \lambda_2 &= 0 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \\ \lambda_1 + \lambda_3 &= 0 \quad 1 \quad 1 \quad 2 \quad 2 \quad 1 \\ \lambda_2 + \lambda_3 &= 0 \quad 0 \quad 1 \quad 2 \quad 2 \quad 1 \end{aligned}$	—
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

[illegible]

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00922 (now)	153	—	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\right)\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\right)=\left(\begin{array}{c}\left(\begin{array}{ccc}h\left(x_{156}\right)&h\left(x_{164},\bar{x}_{256}\right)&h\left(x_{145},\bar{x}_{356}\right)\\h\left(x_{256},\bar{x}_{164}\right)&h\left(x_{264}\right)&h\left(x_{245},\bar{x}_{364}\right)\\h\left(x_{356},\bar{x}_{145}\right)&h\left(x_{364},\bar{x}_{245}\right)&h\left(x_{345}\right)\end{array}\right)\\\left(\begin{array}{ccc}h\left(x_{423}\right)&h\left(x_{431},\bar{x}_{523}\right)&h\left(x_{412},\bar{x}_{623}\right)\\h\left(x_{523},\bar{x}_{431}\right)&h\left(x_{531}\right)&h\left(x_{512},\bar{x}_{631}\right)\\h\left(x_{623},\bar{x}_{412}\right)&h\left(x_{631},\bar{x}_{512}\right)&h\left(x_{612}\right)\end{array}\right)\\\left(\begin{array}{cc}h\left(x_{123}\right)&h\left(x_{456}\right)\end{array}\right)\end{array}\right),$	<i>(not rendered)</i>	—
basis: inferred / ink					
0902.0431_E01032 (was)	173	■ 0.022	$\begin{aligned}&\frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1\quad 2\quad 3\quad 3\quad 2\quad 2\end{aligned}\quad 1\quad \square=\frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2+\lambda_3\right)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1\quad 2\quad 3\quad 3\quad 2\quad 2\quad 1\quad 1\quad 1\quad 1.$	<i>(not rendered)</i>	—
0902.0431_E01032 (now)	173	—	$\begin{aligned}\kappa\pi H&=\tfrac{1}{2}(-\lambda_0+\lambda_1-\lambda_2+\lambda_3)H_0+\tfrac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)H_1\\&\quad +\tfrac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)H_2+\tfrac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)H_3,\end{aligned}$		—
basis: inferred / ink					
0902.0431_E01059 (was)	180	■ 0.093	$\begin{aligned}&\&=\left\{\alpha\in E_8\mid\tau\tilde\alpha=\alpha\tau\tilde\lambda\right\}\\&=\left\{\alpha\in E_7^C\mid\tau\tilde\lambda\alpha=\alpha\tau\tilde\lambda\right\}\text{ (Proposition 5.7.1)}\\&=\left\{\alpha\in E_7^C\mid\tau\lambda\alpha=\alpha\tau\lambda\right\}\text{ (by the correspondence to Proposition 5.7.1)}\\&=E_7\text{ (Lemma 4.3.3.(4)).}\end{aligned}$		—
0902.0431_E01059 (now)	180	—	$\begin{aligned}\kappa\pi H&=\tfrac{1}{2}(-\lambda_0+\lambda_1-\lambda_2+\lambda_3)H_0+\tfrac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)H_1\\&\quad +\tfrac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)H_2+\tfrac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)H_3,\end{aligned}$		—
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01117 (was)	189	■0.108	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = \left(\left(X_1, Y_1 \right) + \left(X_2, Y_2 \right) + \left(X_3, Y_3 \right) \right) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) \\ (X_2, Y_1) & (X_2, Y_2) \\ (X_3, Y_1) & (X_2, Y_3) \\ X \vee \boldsymbol{Y} & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X}Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	—	
0902.0431_E01117 (now)	189	—	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = \left(\left(X_1, Y_1 \right) + \left(X_2, Y_2 \right) + \left(X_3, Y_3 \right) \right) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \wedge \boldsymbol{Y} = \begin{pmatrix} X_2 \wedge Y_3 - Y_2 \wedge X_3 \\ X_3 \wedge Y_1 - Y_3 \wedge X_1 \\ X_1 \wedge Y_2 - Y_1 \wedge X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	—	

basis: inferred / ink

Conf.

MathPix confidence:

■

≥0.9

■

0.5-0.9

■

<0.5

—no value

Residual

render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ1187 (was)	200	■ 0.183	$\begin{aligned} \&=(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{a}, *(\boldsymbol{b} \wedge \boldsymbol{x})) - \\ &\left(\boldsymbol{a}_1, \boldsymbol{w}\right) \\ &\left(\boldsymbol{w} \wedge \boldsymbol{a}_2, *(\boldsymbol{b} \wedge \boldsymbol{x})\right) \&+ \\ &\left(\boldsymbol{a}_2, \boldsymbol{w}\right) \\ &\left(\boldsymbol{w} \wedge \boldsymbol{a}_1, *(\boldsymbol{b} \wedge \boldsymbol{x})\right) - \\ &\frac{3}{5}*(\boldsymbol{a} \wedge \boldsymbol{b}), \\ &\boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \&= \\ &\frac{2}{5}*(\boldsymbol{a} \wedge \boldsymbol{b}), \\ &\boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) - \\ &\left(\boldsymbol{a}_1, \boldsymbol{w}\right) \\ &\left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_2, \right. \\ &\left. \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5\right) \&+\left(\boldsymbol{a}_2, \right. \\ &\left. \boldsymbol{w}\right)\left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \right. \\ &\left. \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5\right) \&((\\ &\boldsymbol{b} \times \text{⊞} \text{⊞} - \text{⊞} \text{⊞} - \text{zh20} \\ &\&=(\boldsymbol{a} \wedge \boldsymbol{x})) \\ &\boldsymbol{v}, \boldsymbol{w}) \&=(\boldsymbol{v}, \\ &\boldsymbol{w})(*(\boldsymbol{a} \wedge \boldsymbol{b}), \\ &\boldsymbol{v})+\left(\boldsymbol{a}_1, \boldsymbol{w}\right) \\ &\right)\left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \right. \\ &\left. \boldsymbol{a}_2, \right. \\ &\left. \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5\right) \&-\left(\boldsymbol{a}_2, \right. \\ &\left. \boldsymbol{w}\right)\left(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \right. \\ &\left. \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5\right) -\frac{3}{5}*(\\ &\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\\ &\boldsymbol{v}, \boldsymbol{w}) . \end{aligned}$	(not rendered)	—
0902.0431_EQ1187 (now)	200	—	$\begin{aligned} H \&= \nu \Big(\sum_{k=0}^3 \lambda_k H_k \\ &\&= \tfrac{1}{2}\big(\lambda_0(-H_0 + H_1 - H_2 + H_3) + \lambda_1(-H_0 + H_1 + H_2 - H_3) \\ &\quad + \lambda_2(H_0 + H_1 + H_2 + H_3) + \lambda_3(-H_0 - H_1 + H_2 + H_3)\big) \\ &\&= \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_0 + \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_1 \\ &\quad + \tfrac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_2 + \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_3, \end{aligned}$	$\begin{aligned} H &= \nu \Big(\sum_{k=0}^3 \lambda_k H_k \Big) \\ &= \tfrac{1}{2}(\lambda_0(-H_0 + H_1 - H_2 + H_3) + \lambda_1(-H_0 + H_1 + H_2 - H_3) \\ &\quad + \lambda_2(H_0 + H_1 + H_2 + H_3) + \lambda_3(-H_0 - H_1 + H_2 + H_3)) \\ &= \tfrac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3)H_0 + \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &\quad + \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \tfrac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \end{aligned}$	—
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E08802	134	■ 0.338 •	$\left(\begin{array}{c} \cos a_1 \frac{\sin a_2 }{ a_2 } \frac{\sin a_3 }{ a_3 } \\ \frac{\sin a_1 }{ a_1 } \cos a_2 \cos a_3 \\ 0 \end{array}\right) \left(\begin{array}{ccc} \cos a_1 \frac{\sin a_2 }{ a_2 } \frac{\sin a_3 }{ a_3 } & 0 & 0 \\ 0 & a_1 \frac{\sin a_1 }{ a_1 } \cos a_2 \frac{\sin a_3 }{ a_3 } & 0 \\ 0 & 0 & a_1 \frac{\sin a_1 }{ a_1 } a_2 \frac{\sin a_2 }{ a_2 } \cos a_3 \end{array}\right)$	$\left(\begin{array}{ccc} \cos a_1 \frac{\sin a_2 }{ a_2 } \frac{\sin a_3 }{ a_3 } & 0 & 0 \\ 0 & a_1 \frac{\sin a_1 }{ a_1 } \cos a_2 \frac{\sin a_3 }{ a_3 } & 0 \\ 0 & 0 & a_1 \frac{\sin a_1 }{ a_1 } a_2 \frac{\sin a_2 }{ a_2 } \cos a_3 \end{array}\right)$	$\left(\begin{array}{ccc} \cos a_1 \frac{\sin a_2 }{ a_2 } \frac{\sin a_3 }{ a_3 } & 0 & 0 \\ 0 & a_1 \frac{\sin a_1 }{ a_1 } \cos a_2 \frac{\sin a_3 }{ a_3 } & 0 \\ 0 & 0 & a_1 \frac{\sin a_1 }{ a_1 } a_2 \frac{\sin a_2 }{ a_2 } \cos a_3 \end{array}\right)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00939	156	■0.471 ●C +23	$\left.\begin{array}{l} \xrightarrow{\quad} \left(\begin{array}{ccc} h\left(b_{11}-c_{11}-\frac{\nu}{3}\right) & -h\left(c_{12}, b_{12}\right) & -h\left(c_{13}, b_{13}\right) \\ -h\left(\bar{b}_{12}, \bar{c}_{12}\right) & 0 & 0 \\ -h\left(\bar{b}_{13}, \bar{c}_{13}\right) & 0 & 0 \end{array} \right) \\ = \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & h\left(l_{33}\right) & -h\left(l_{23}, \bar{l}_{32}\right) \\ 0 & -h\left(l_{32}, \bar{l}_{23}\right) & h\left(l_{22}\right) \end{array} \right) \\ 0 \\ -h\left(\bar{l}_{11}\right) \end{array} \right) .$	$\xrightarrow{f^{-1}} \left(\begin{array}{c} \left(\begin{array}{ccc} h\left(b_{11}-c_{11}-\frac{\nu}{3}\right) & -h\left(c_{12}, b_{12}\right) & -h\left(c_{13}, b_{13}\right) \\ -h\left(\bar{b}_{12}, \bar{c}_{12}\right) & 0 & 0 \\ -h\left(\bar{b}_{13}, \bar{c}_{13}\right) & 0 & 0 \end{array} \right) \\ \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & h\left(l_{33}\right) & -h\left(l_{23}, \bar{l}_{32}\right) \\ 0 & -h\left(l_{32}, \bar{l}_{23}\right) & h\left(l_{22}\right) \end{array} \right) \\ 0 \\ -h\left(\bar{l}_{11}\right) \end{array} \right)$ $= \left(\begin{array}{c} \phi_{\boldsymbol{C}}(B, C) E_1 + \frac{i\nu e_1}{3} E_1 \\ 2h(L) \times E_1 \\ 0 \\ -(\tau h(L), E_1) \end{array} \right) = \Phi(\phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i\nu e_1) \left(\begin{array}{c} E_1 \\ 0 \\ 0 \\ 0 \end{array} \right) .$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 —no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00696	115	0.535	$\begin{array}{c} \backslash\begin{array}{cccc} \Phi_{-3}=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \kappa & 0 \\ 0 & 0 & C & 0 \end{array}\right), & \Phi_3=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_{-2}=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & d & 0 & 0 \end{array}\right), & \Phi_2=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ c & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_{-1}=\left(\begin{array}{cccc} 0 & l & 0 & 0 \\ 0 & 0 & B & 0 \\ 0 & 0 & 0 & 0 \\ b & 0 & 0 & 0 \end{array}\right), & \Phi_1=\left(\begin{array}{cccc} 0 & 0 & 0 & A \\ k & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_0=\left(\begin{array}{cccc} g & 0 & 0 & 0 \\ 0 & h & 0 & 0 \\ 0 & 0 & \nu & 0 \\ 0 & 0 & 0 & \mu \end{array}\right). \end{array}$	$\begin{array}{c} \Phi_{-3}=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \kappa & 0 \\ 0 & 0 & C & 0 \end{array}\right), & \Phi_3=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_{-2}=\left(\begin{array}{cccc} 0 & 0 & C & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & d & 0 & 0 \end{array}\right), & \Phi_2=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & D \\ c & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_{-1}=\left(\begin{array}{cccc} 0 & l & 0 & 0 \\ 0 & 0 & B & 0 \\ 0 & 0 & 0 & 0 \\ b & 0 & 0 & 0 \end{array}\right), & \Phi_1=\left(\begin{array}{cccc} 0 & 0 & 0 & A \\ k & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_0=\left(\begin{array}{cccc} g & 0 & 0 & 0 \\ 0 & h & 0 & 0 \\ 0 & 0 & \nu & 0 \\ 0 & 0 & 0 & \mu \end{array}\right). \end{array}$	
0902.0431_E00037	11	0.555 C -48	$\begin{array}{c} \backslash\begin{array}{lll} 1 \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{array}\right), & \kappa \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array}\right), & \pi \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array}\right), \\ \nu \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array}\right), & \nu^2 \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array}\right), & \nu\pi \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array}\right) \end{array}$	$\begin{array}{c} 1 \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 1 & 2 & 3 \end{array}\right), & \kappa \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array}\right), & \pi \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array}\right), \\ \nu \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array}\right), & \nu^2 \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array}\right), & \nu\pi \rightarrow \left(\begin{array}{lll} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array}\right) \end{array}$	
Conf. MathPix confidence: 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					